

1-Bit Full Subtractor Using Dataflow Modeling

Aim

To design and verify a 1-bit full subtractor in Verilog using dataflow modeling.

Code

```
module full_subtractor(  
    input A,  
    input B,  
    input Bin,  
    output Difference,  
    output Borrow  
);  
  
assign Difference = A ^ B ^ Bin;  
  
assign Borrow = ((~A) & B) | ((~A) & Bin) | (B & Bin);  
  
endmodule
```

Truth Table / Observation

A	B	Bin	Difference	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Observation: All 8 input combinations were simulated and matched the expected full-subtractor behavior. The simulation generated `full_subtractor.vcd` and completed at 80 ns.

Conclusion

The 1-bit full subtractor was successfully implemented using Verilog dataflow modeling. The XOR network produces the difference, and the borrow equation correctly identifies when subtraction requires a borrow.